Department of Economics

MSc Extended Essay Cover Sheet

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COURSE CODE	E	C	4	2	3

TITLE

The Heterogeneous Impact of COVID-19 Income Support Programs on Household Consumption in Australia

WORD COUNT	5	9	9	2

Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text).*

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The Heterogeneous Impact of COVID-19 Income Support Programs on Household Consumption in Australia

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Abstract

In response to the COVID-19 pandemic, the Australian government introduced several large-scale income support programs that have been subjected to heated discussion. This includes one of the most expensive job retention programs called the Job Keeper. It is crucial for policymakers to evaluate the effects of the programs in retrospect. This essay studies the heterogeneous effect of government policies on household consumption and financial stress experiences using individual-level longitudinal survey data.

The Job Keeper program offered a flat payment to eligible workers irrespective of their regular income before the pandemic. This meant that the payment exceeded the regular income for low-income employees, while only replacing a portion of the regular income for those with higher-earning jobs. The essay shows that the program increased consumption for workers with regular income below the Job Keeper payment, and the increase was mostly driven by discretionary goods spending. In addition, low-income workers who received the payment were less likely to experience financial difficulties. In contrast, the program had no impact on the consumption of higher-income workers. The findings contribute to several aspects of the discussion on the design of government policies.

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Motivation and Background

The COVID-19 pandemic has had a major long-lasting impact on the economy and financial system. In response to the pandemic, many countries including Australia introduced various income support measures to help people get through financial hardships and cushion the economic impact caused by the pandemic. The COVID-19 pandemic was unprecedented, and authorities were forced to make policy decisions quickly without detailed planning.

It is inherently important for policymakers to understand and evaluate the effects of large-scale government support measures in retrospect. Establishing research-based evidence will guide to design and refine future policies. The COVID-19 restrictions caused large and unexpected changes in household spending behaviour overall (Bishop, Boulter, & Rosewall,2022). The negative shock of the pandemic was felt unevenly across individuals and families with different characteristics. This essay examines the heterogeneous effect of government policies on household consumption and financial stress experiences using individual-level longitudinal survey data.

The Australian government introduced a new job retention program called Job Keeper (JK) and an Early Release of Superannuation (ERS) scheme in response to the pandemic. Both programs have been subjected to heated discussion. The essay has a stronger focus on the effect of the JK program as it is the most expensive job subsidy program in Australia's history and perhaps the one that received the most discussion. The JK program was shown to be effective in achieving its primary goal of maintaining employer-employee connections. Bishop and Day (2020) found that 1 out of 5 employees would have lost their jobs without the JK payment. Borland & Hunt (2021) showed that the JK program helped to maintain employer-employee connections by encouraging temporary layoffs than permanent layoffs. But it has been less

clear about its effect on individual workers' financial situations. To the best of my knowledge, this is the first study that examines the heterogeneous effect of the JK subsidy on workers' spending and financial stress experience using survey data. The essay provides evidence that the program benefited low-income workers more.

Job Keeper Background

The primary goal of the JK program was to support businesses and workers who were negatively affected by the pandemic and maintain formal employer-employee relationships. It was introduced in March 2020 and ended in March 2021. Over a quarter of the Australian employees were receiving the JK payment at its peak time. The program cost around A$90 billion over its entire duration and was the most expensive income support program in Australia's history.

A unique feature of the JK program was that it offered a flat payment to all employees irrespective of their regular income before the pandemic. In contrast, similar programs in most developed countries offered payments indexed to workers' regular wages including the UK's Furlough scheme. The JK program was set at A$1,500 (around £850) per fortnight in the initial phase that lasted for six months. Then the payment was progressively reduced.

Not all employees whose incomes were negatively affected by the pandemic were eligible to receive the JK payment. Their employers must have experienced a loss in revenue beyond certain thresholds (The Federal Register of Legislation, 2021). Casual workers had to be employed on 1 March 2020 for at least 12 months to be eligible. Sole traders and self-employed people were eligible.

Early Release of Superannuation Scheme Background

The objective of the ERS scheme was to provide financial relief to individuals who might be experiencing financial hardship. To be eligible, individuals must either lose at least some income due to the pandemic or be already receiving some form of income support payments. Under the scheme, eligible people could withdraw from their superannuation accounts twice, each in the first and second half of 2020 (Up to A$10,000 each time). The amounts released under the scheme were not subject to personal income tax. Over 10 per cent of the Australian population withdrew their superannuation early under the scheme. Although the scheme was intended to provide financial relief to people who were struggling financially, people accessed the super early for other reasons including for precautionary reasons or paying off their mortgages.

Although both programs aimed at providing financial support for businesses and households, several distinctions might lead people to respond to the programs differently. Firstly, the ERS scheme was financed through individuals' own superannuation account balances. Eligible individuals chose whether they wanted to access their super early. Eligible people may not have had any superannuation balances in their accounts. Furthermore, the ERS scheme allowed access to a large lump sum of money, whereas the JK program offered a relatively small amount of payment on a fortnightly basis.

Data and Definitions

The essay relies on the Household, Income and Labor Dynamics in Australia (HILDA) Survey. It is Australia's most comprehensive longitudinal survey data and has been used extensively in academic research. HILDA is an annual survey that tracks a group of

representative individuals living in Australia since 2001. The latest 2021 data capture the first year that people in Australia were largely able to resume normal activities. The first case of COVID-19 was confirmed in January 2020 in Australia. Most domestic travel restrictions were lifted in late 2020, and people began to return to their normal lives.

The survey collects detailed and rich information on household and individual characteristics on a variety of subjects including labour market status, income, education and consumption. Every four years, HILDA includes a wealth module that collects household assets and debt data. The wealth data are up to 2018. The longitudinal nature of the data allows us to track an individual's behaviour over time, enabling us to study the consequences of certain life events such as whether receiving a wage subsidy impacts people's behaviour in later years.

There are some variables worth highlighting. The 2020 survey features a special COVID-19 pandemic theme. It includes a range of COVID-19-related questions to understand how the pandemic impacted the lives of individuals. For example, the survey asks whether an individual received government income support payment, temporarily or permanently lost their jobs, reduced or increased their working hours, worked from home more, took on additional debts etc. The survey explicitly asks if the events happened as a result of the pandemic so that respondents would not misinterpret it as usual events. The share of workers receiving the JK payments was around a quarter in the survey data, close to the official statistics published by the Australia Treasury. In addition to the COVID-19-related financial difficulties in 2020, the

survey collects financial stress-related questions every year such as whether an individual was unable to pay rent or mortgage on time or went without meals due to financial difficulties.[1]

The survey contains detailed information on household consumption of non-durable goods each year. Durable goods data are not available for the period relevant to the study. I created several categories for spending including total non-nondurable goods, food spending, essential goods and discretionary goods.[2]

Characteristics of Job Subsidy Recipients

The JK recipients were different from the recipients of the regular unemployment subsidy called Jobseeker (JS). The JK recipients were employed before the pandemic, whereas

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[1] The survey asks if an individual experiences any three or more of the following events due to financial difficulties: could not pay electricity, gas or telephone bills on time; could not pay rent or mortgage on time; asked for financial help from friends or family; unable to heat home; went without meals; pawned or sold something; asked for help from welfare/community organisations. (Melbourne Institute, 2021)

[2] Total non-durable spending is the sum of all items collected by the survey. This includes groceries, meals eaten out, alcohol and tobacco, transport, holiday travel, clothing and footwear, motor vehicle fuel and maintenance, home repairs, renovation and maintenance healthcare fees and products, utilities, telecommunications, education fees and insurance; Food spending includes groceries and meals eaten out. I created this measure because past research found that food spending is least likely to suffer from measurement error in survey data as it is easy for people to recall how much they spend on groceries and eating out than spendings on other items; Essential goods include groceries, utilities and bills; Discretionary goods consist of spending on alcohol, tobacco, meals eaten out, clothing and home improvements. I did not include spending on activities that were severely limited due to lockdown restrictions such as transportation and travel. Whether I include these spendings in the discretionary goods does not alter the overall observations much. (Melbourne Institute, 2021)

the JS recipients were unemployed. The data show that the JK receivers were more educated, less likely to be in financial stress, earned more, less likely to have pre-existing long-term health conditions, and were more likely to own their own home rather than renting compared to JS receivers. Overall, the JK receivers tended to be more well-off than the JS recipients.

Table 1: Characteristics of Job Subsidy Recipients

Prior to the COVID-19 pandemic (2019), average

	Workers who did not receive Job Keeper	Workers who received Job Keeper	Job Seeker
Annual income ($A)	133926	134706	62980
High-risk industry* (%)	35.8	51.2	48.8
Tertiary education (%)	36.1	28.5	10.1
Age (years)	40	39	40
Renter (%)	30	31.6	66.1
Mortgagor (%)	47.1	46.2	10.7
Long-term health issues (%)	16.3	16.2	49.1
Has young children (%)	37.6	36	28.8
Financial stress (%)	4.5	4.9	27.4
Hand-to-mouth* (%)	13.9	13.8	46.7

* High-risk industries are industries that typically require working in proximity to people. This includes arts and recreation, trade, construction, manufacturing and rental.

**Hand-to-mouth refer to people who are liquidity-constrained; calculated with 2018 wealth data.

Many people criticized that the JK program was too expensive, and it placed a heavy burden on public debt. The fact that the JK recipients were better off than JS recipients raised the question that whether the JK program was paying too much to people who might not need the money as much as the JS recipients. Although the Australian government temporarily boosted the JS payment (almost double), it was still less than the JK payment.

Methodology

I used a difference-in-difference model to identify the causal effect of the program. The basic model is specified as:

$$Outcome_{it} = \alpha_i + \beta Treatment_i * Post_i + \delta Treatment_i + \phi Post_t + \gamma Controls_{it} + \varepsilon_{it}$$

$Treatment_i$ is a dummy variable that equals to one for the treatment group and zero otherwise. $Post_t$ is a time dummy variable that equals to one after the policy was introduced. The timing of the policy change can be considered as exogenous as most people did not predict the pandemic and its consequences. Alternatively, I used $\sum_t \phi_t Year_t$ instead of $Post_t$ to control for year fixed effects. $Controls_{it}$ include a range of control variables that could affect individuals' financial situations including income, age, whether they had young children as well as dummy variables for major life events that happened recently such as separated from partners or getting pregnant etc. Individual fixed effects were included to account for time-invariant individual characteristics. The individual fixed effects would absorb industry fixed effects to the extent that people do not change industries. It is uncommon for people to shift their career pathways frequently, so the industry-fixed effects would be largely captured by the individual fixed effects. The coefficient of the interaction term between the treatment group and post-treatment period dummies β will pick up the treatment effect.

To understand how the effect evolved over time, I estimated a dynamic DID model with time-fixed effects. A generic model is specified as:

$$Outcome_{it} = \alpha_i + \sum_t \beta_t (Treatment_i \times Year_t) + \sum_i \delta_i Treatment_i + \sum_t \phi_t Year_t + \gamma Controls_{it} + \varepsilon_{it}$$

$Year_t$ represent time dummy variables for each year. The base year is 2019, the year before the introduction of the COVID-19-related policies.

Treatment and Control Groups

Job Keeper

I defined the treatment group as people who received the Job Keeper subsidy, and the control group as those who did not receive the payment. Only people who were employed before the pandemic were included to make the treatment and control groups more comparable. Employers' profitability had to fall below certain thresholds for their employees to be eligible. This is out of workers' control, so it can be considered exogenous.

The identification strategy relies on the assumption that the differences in the outcome of workers who received and did not receive the subsidy would have evolved similarly had it not been for the JK payment. I would be more confident about the parallel assumptions if the two groups were very similar. As shown in the descriptive statistics Table 1, the treatment and control groups were similar in many aspects before the pandemic. They had similar income levels and were of similar age on average. Moreover, there were little differences between the shares of people who were renting, had mortgage debts, had long-term health issues, had young children and were liquidity-constrained between the two groups. These were the key characteristics that could affect consumption and financial stress experience. The key distinctions between the treatment and control groups were the industry of work and education level. The JK recipients were much more likely to work in industries that were heavily adversely impacted by the pandemic and were less likely to hold tertiary education qualifications. This could be because professions that require tertiary education were generally more able to work from home and therefore less likely to qualify for the JK payments. There

could be some unobserved differences between the two groups that affected the outcome. For example, college graduates might be more financially literate, so they could be better at managing their money.

Given that the main factors that could affect financial circumstances such as income, housing status, mortgage debts etc. between the treatment and control groups were similar, it is fair to assume that the consumption and financial stress experiences of the two groups would have evolved similarly if it were not for the JK subsidy.

I considered two alternative approaches that would make the two groups very similar in theory. The first is a fuzzy regression discontinuity framework that exploits the job tenure requirement for casual employers. I could restrict the sample to casual workers who worked just above and below 12 months to ensure that the people in the sample are very similar. The treatment group would be those who passed the job tenure requirement and those who did not would be the control group. The survey data reveal that a small share of casual employers with less than one year of job tenure reported that they received the JK payment. This suggests that the requirements were not strictly enforced. A fuzzy regression discontinuity design would resolve this issue by using whether a casual employee's job tenure exceeded the 12 months threshold as an instrumental variable. A huge advantage of this strategy is that the treatment would be almost as good as randomly assigned. There would be no systematic differences between the treatment and control groups, so any differences in outcome can be attributed to the effect of the JK program. This approach would allow me to identify the causal effect of the JK program for casual workers who worked for around one year. The second approach uses the fact that some ineligible casual workers received the payment. I could restrict the sample

to ineligible casual workers, using those who received the JK payment as the treatment group and others as the control group. The two groups would be very similar.

The treatment and control groups identified with the two alternative approaches are very likely to satisfy the parallel trend assumption. But the locality of the estimate could be a limitation. The effect on casual employees may or may not be generalized to other types of employees. Permanent full-time employees might respond differently as they tend to be older and were more financially stable than casual workers. Unfortunately, both approaches required me to restrict the sample to a very narrow range of individuals and the sample size is too small to yield any meaningful results.

Early Release of Superannuation Scheme

I defined the treatment group as people who withdrew early superannuation early. The control group consisted of people who were eligible for the scheme but had little to no money in their superannuation. People are allowed to withdraw their super early under some special circumstances during normal times. People who had no money in their accounts might have cleared out their super for other reasons such as using it to purchase their first home. The data on superannuation account balances were available every four years and the latest data were from 2018. I identified the control group as people who did not have any money in their superannuation accounts in 2018. Although these workers would have contributed some money to their super accounts by 2020, they would not be able to accumulate much.

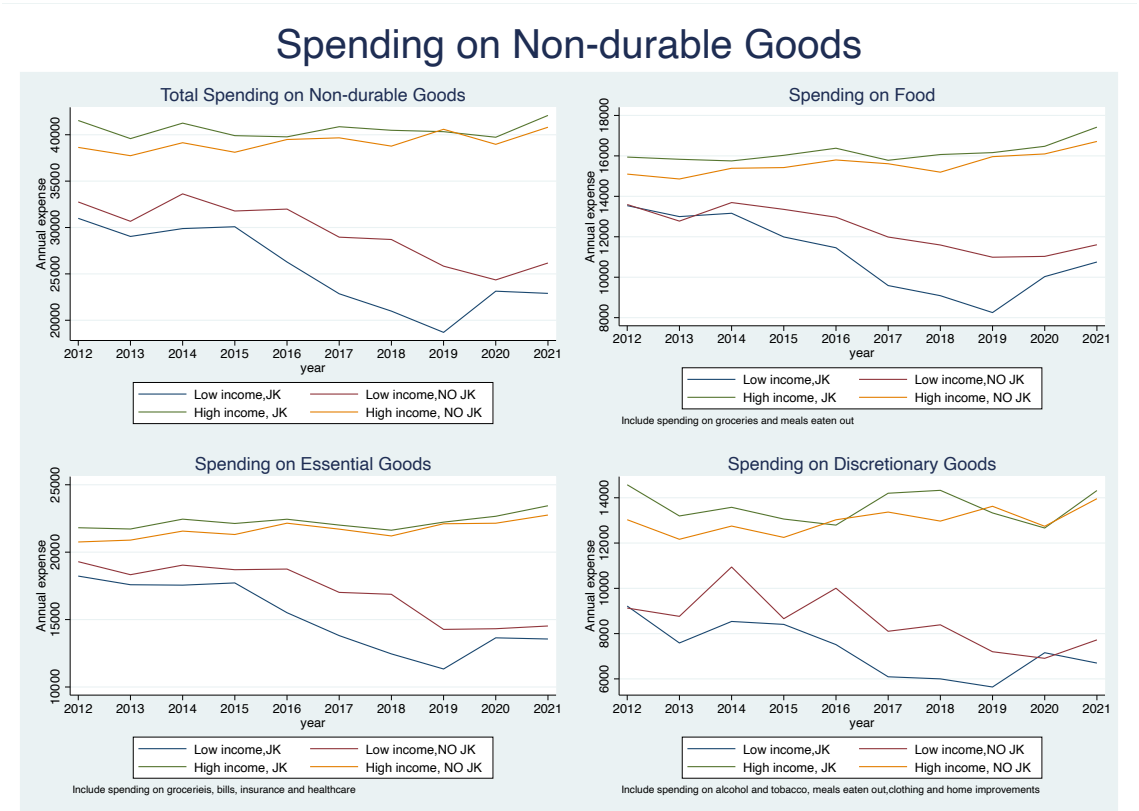
Impact of the JK Payments on Consumption

Descriptive evidence

The flat payment structure of the JK program meant that the payment exceeded the regular income for low-income employees, while only replacing a proportion of the regular income for those with higher-earning jobs. So, the subsidy payment would have different implications for people with varying income levels.

Aggregate data would obscure the distributional effects. To uncover the heterogeneity effect, I split the sample into low and high-income workers based on whether their income before the pandemic exceeded the Job Keeper payment rate of A$1500 per fortnight.

Graph 1

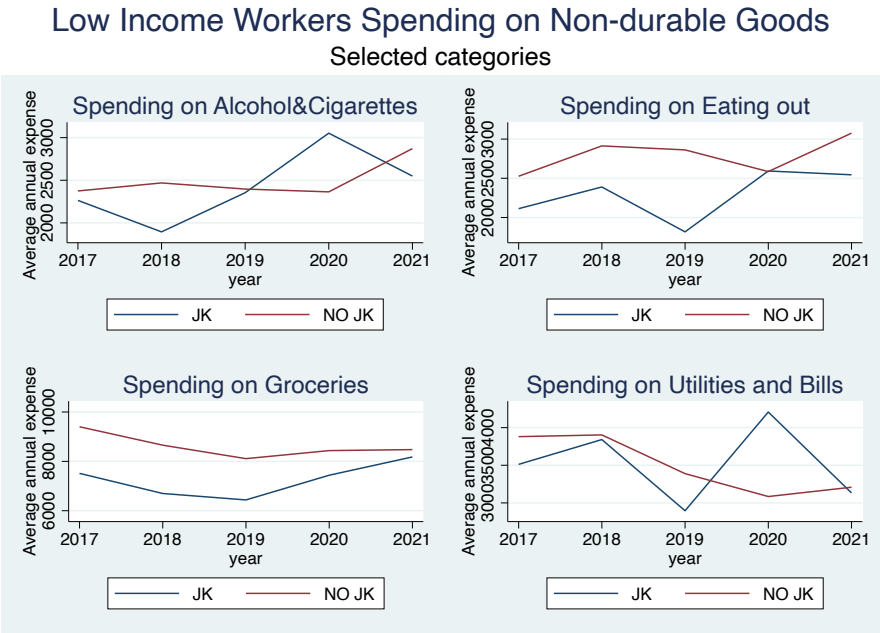


Graph 1 plots average spending by income levels and whether they received the JK payment. There are a few facts worth highlighting. Spending for high-income households had been relatively steady over the past decade and the trends for the JK and non-JK receivers among high-income workers were similar. In contrast, the spending patterns for low-income households were different for those who received and did not receive the JK payments. On average, the spending of low-income workers was trending downwards in the lead-up to the pandemic. The trends between the JK and non-JK workers were somewhat parallel. Interestingly, the trends started to differ after the JK program was introduced in 2020 for low-income workers. Low-income workers who received the JK payments increased their spending on both essential and discretionary goods while the spending of the non-JK receivers was little changed from 2019 to 2021, with a slight decrease in spending on discretionary goods in 2020. This hints at the possibility that the JK payment had a positive impact on consumption for low-income workers.

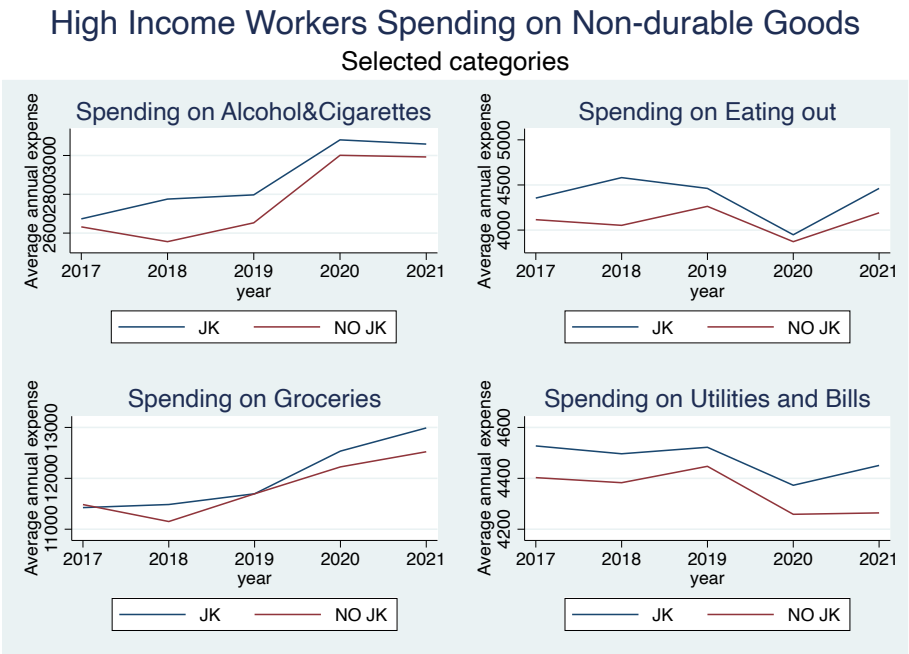
Graphs 2 and 3 take a closer examination of the types of spending that drove the changes for low and high-income workers. It appears that changes in consumption patterns were most volatile for the low-income JK recipients. The Low-income JK receivers increased their spending on more discretionary items such as alcohol, cigarettes and eating out compared to their counterpart. This indicates that the increase in income for low-income workers might have encouraged some to splurge on things that they would not normally buy. The low-income JK recipients also spent a lot more on utilities and bills on average, suggesting that some of the JK subsidy payments might have been used to pay for bills. The spending patterns for the JK and non-JK recipients for high-income workers were relatively similar. They reduced their spending on eating out in 2020, possibly due to the lockdown restrictions. High-income

households might have replaced their spending on eating out by increasing spending on alcohol, cigarettes and groceries in 2020 and 2021.

Graph 2



Graph 3



Regression evidence

To identify the causal effect of the JK payments on spending. I estimated a series of DID models and gradually added in more controls. A generic model is specified as follows.

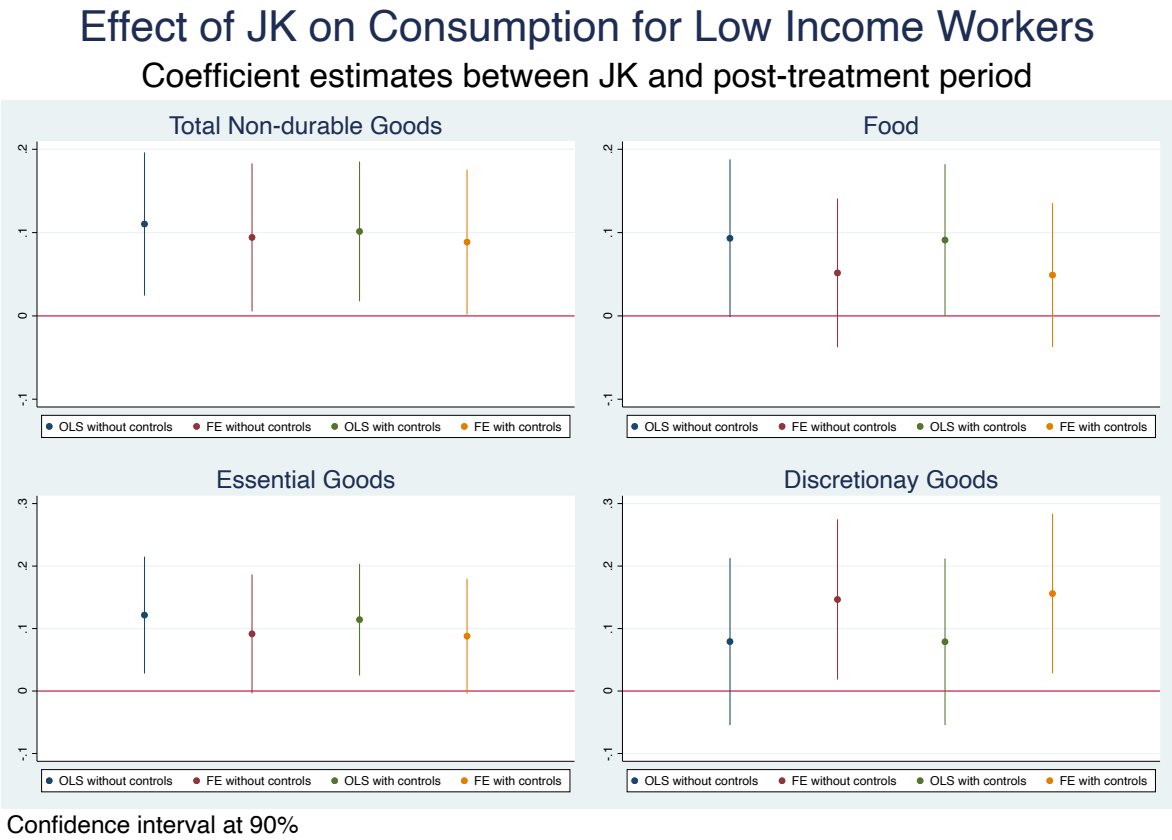
$$Log\ of\ spending_{it} = \alpha_i + \beta Treatment_i * Post_i + \delta Treatment_i + \phi Post_t + \gamma Controls_{it} + \varepsilon_{it}$$

I ran the regressions on low-income households and all households. For validity checks, I also included dummy variables for workers' locations to account for the uneven spread of COVID-19 and the degree of restrictions across regions. Using $\sum_t \phi_t Year_t$ instead of $Post_t$ to control for year fixed effects gave similar results. I used clustered standard errors to account for serial correlations.

The coefficient estimates in the sample with all households were insignificant. But many estimates became significant when the sample was restricted to low-income households. This shows that the aggregate data would obscure the heterogenous effect of the JK program on households and the effect was more significant for low-income people.

Graph 4 plots the coefficient estimates of the interaction term between the treatment group and the post-treatment period across different model specifications from the low-income household sample. The estimates from the most stringent model with individual fixed effects and time-variant controls suggest that receiving the JK payments increased spending on total non-durable goods by around 9% for the low-income JK receivers, driven by the effect on discretionary goods (around 16%). It does not appear that the JK program affected spending on essential goods. This was probably because essential goods are fundamental to human needs, most people required them in similar quantities.

Graph 4



The model above estimated the average effects of the JK program on consumption. To understand how the effect evolved, I estimated the DID model that allowed for different treatment effects for each year. A generic model is specified as

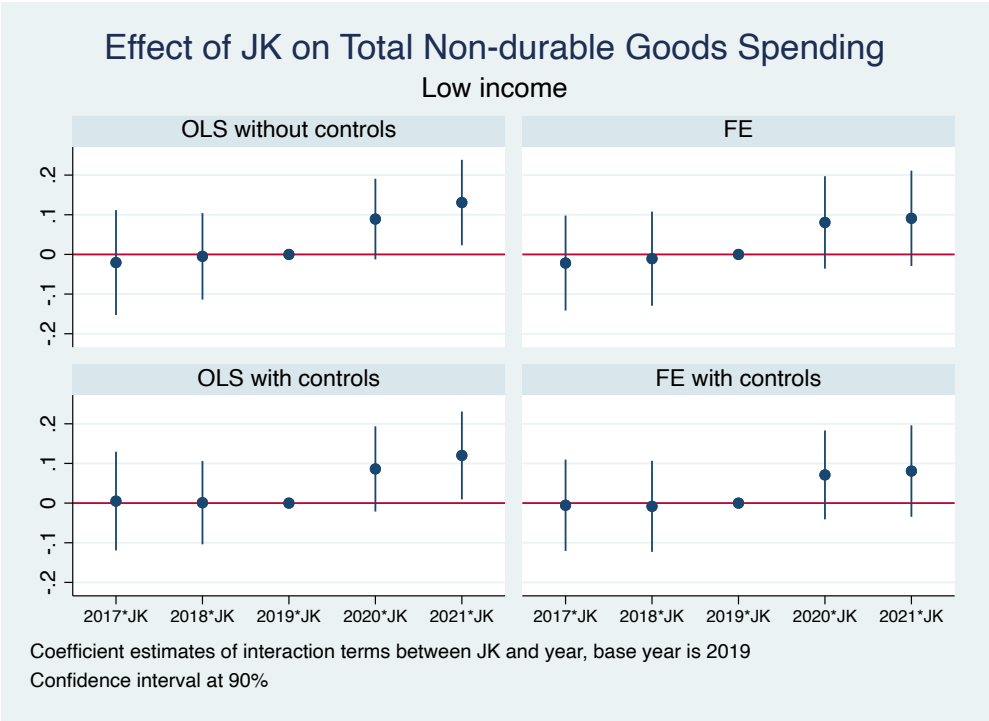
$$Log\ of\ spending_{it} = \alpha_i + \sum_t \beta_t (Treatment_i \times Year_t) + \sum_i \delta_i Treatment_i + \sum_t \phi_t Year_t + \gamma Controls_{it} + \varepsilon_{it}$$

Graphs 5 to 8 show the coefficient estimates of the interaction terms between the treatment group and year dummies, with 2019 as the base year. The model passed the pre-trend test as the interaction terms between the JK and the years before 2019 were insignificant across all specifications. On top of this, the JK and non-JK receivers were similar in many aspects before

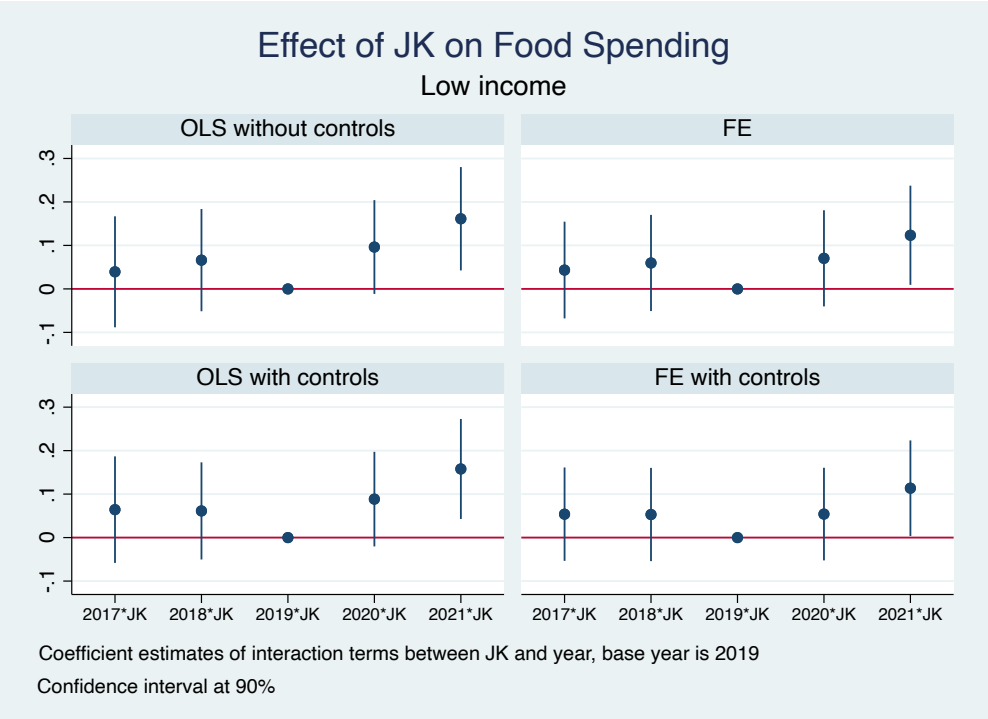
the pandemic as shown in Table 1. Therefore, it is plausible that the parallel trend assumption holds.

The point estimates for the effect of the JK in 2021 were generally larger than the estimates for the effect in 2020. This could reflect that people had more opportunities to spend when the economy opened in 2021. Spending in 2021 may reflect pent-up demand during the lockdown. Some estimates of terms 2020*JK and 2021*JK from the OLS regressions were significant. But the significance disappeared in the strictest model. Splitting the post-treatment period into different years washed away most of the significance that we found previously in the simple DID framework. This could simply be because the effective sample size became too small to detect any significant relationship.

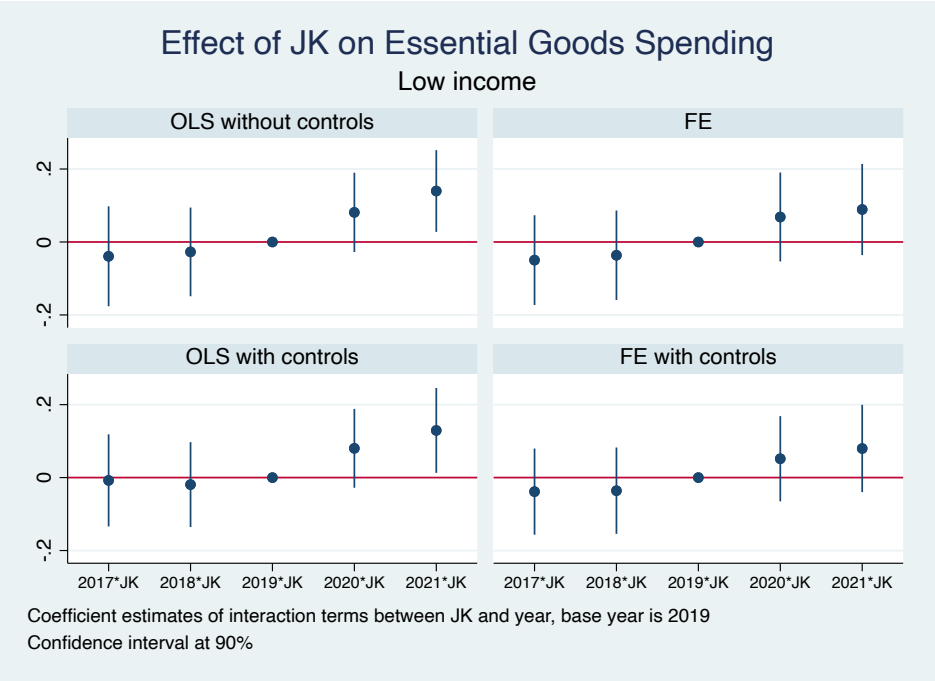
Graph 5



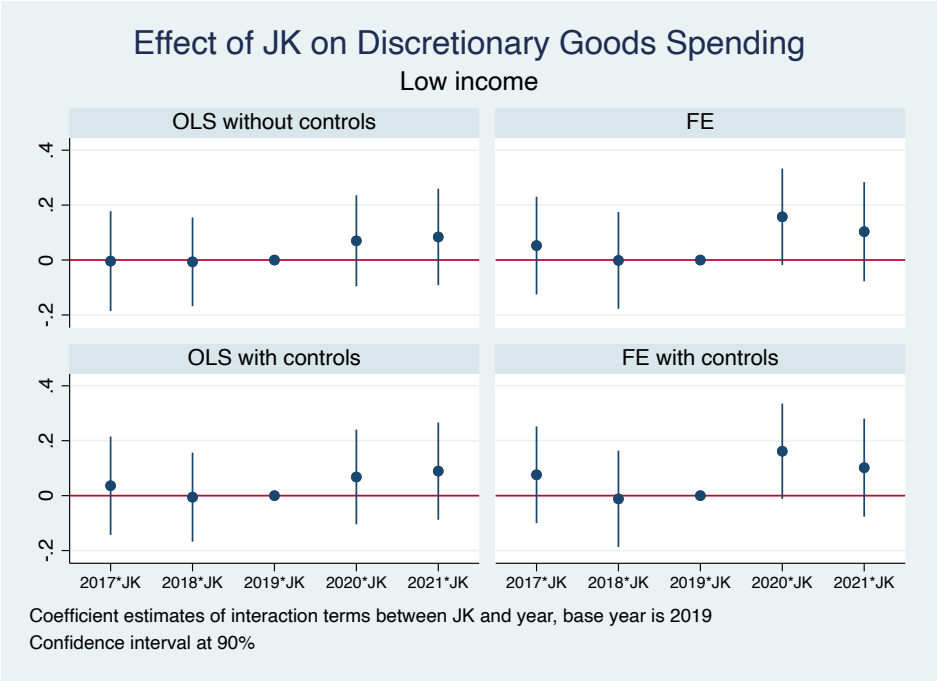
Graph 6



Graph 7



Graph 8



Overall, the model with a pre- vs. post-treatment difference confirmed the hypothesis that the JK program had a heterogenous impact on people with different income levels. It did not affect the spending pattern of high-income households. In contrast, low-income workers who received the JK payment increased their spending on discretionary goods. This finding is consistent with a strand of literature that demonstrates that marginal propensity to consume out of income differs across the income distribution. For example, Parker, Souleles, & Johnson, 2006 found that MPC to consumption out of tax rebates is larger for low-income households.

The fact that the low-income JK receivers increased their spending when they lost work highlights the unique nature of the pandemic shock. Previous literature shows that a typical household reduced their spending by close to 10 per cent during the year of unemployment in Australia in normal times (Penrose and La Cava 2021). The contrast suggests that people might have responded to job loss caused by the pandemic differently. People might be more willing

to spend if they believed that the unemployment caused by the COVID-19 shock was only temporary and that they would go back to work after the recovery. This was particularly true since the JK payment ensured that the employer-employee relationship was maintained. It is also possible that the extra cash that low-income workers received encouraged them to spend on items that were usually outside of their budgets and splurged more on things that they would not have indulged in. This is consistent with the 'windfall effect' in psychology, where people feel more comfortable spending money that they did not work for.

There are a few factors that could obscure the effect of the JK program on consumption. The permanent income hypothesis suggests that individuals tend to smooth lifetime consumption by borrowing and saving. Many people might have built up sizable liquid assets such as cash and deposits in the lead-up to the pandemic. In the event of a negative income shock, they could use their accumulated liquidity buffers to smooth consumption. If this was true, then people who were liquidity constrained (those with little to no liquidity buffers) would respond to the subsidy payments more strongly than people with a sizable liquidity buffer when facing an income shock. To test this hypothesis, I identified liquidity-constrained households following a framework developed by Kaplan, Violante and Weidner (2014). These are households whose liquid assets were very low relative to their income. Household wealth data that are required to identify liquidity-constrained households are collected every four years and the latest data are from 2018. I split the sample into liquidity-constrained households and households that were not liquidity constrained based on the 2018 data. I then ran the same regressions on both samples separately. If the hypothesis that the subsidy affected liquidity-constrained households more was true, then the β estimates from the liquidity-constrained households' sample should be larger. But the estimates were not statistically significant. This

could just be a data issue. A major data limitation is that I could not observe people's liquidity position just before the pandemic. Their liquidity positions might have changed drastically since the latest observed data in 2018.

Another source of bias might come from the spillover effect. For example, it could be that a JK recipient was a close family member or a friend of a non-JK receiver who needed financial help, and the JK recipient used some of the payment to help them out. In this case, the bias would lead us to underestimate the effect of the JK program on consumption because some people who did not receive the payment might have benefited from the program indirectly.

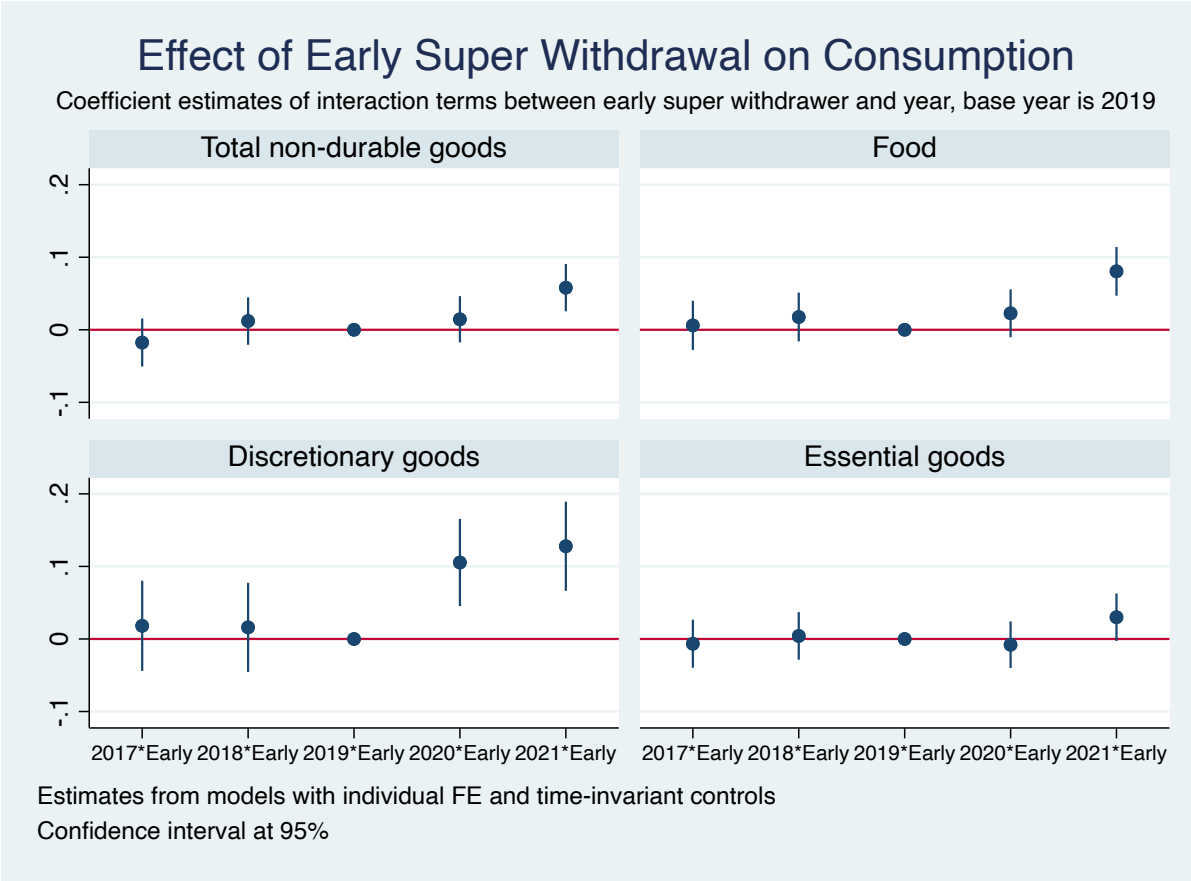
Impact of ERS on Consumption

Next, I estimated a generalised DID model with fixed effects to study the relationship between the ERS scheme and consumption.

$$Log\ of\ spending_{it} = \alpha_i + \sum_t \beta_t (Treatment_i \times Year_t) + \sum_i \delta_i Treatment_i + \sum_t \phi_t Year_t + \gamma Controls_{it} + \varepsilon_{it}$$

The treatment group included people who accessed superannuation early, and the control group consisted of people who were eligible for the program but had little to no money in their super accounts.

Graph 9



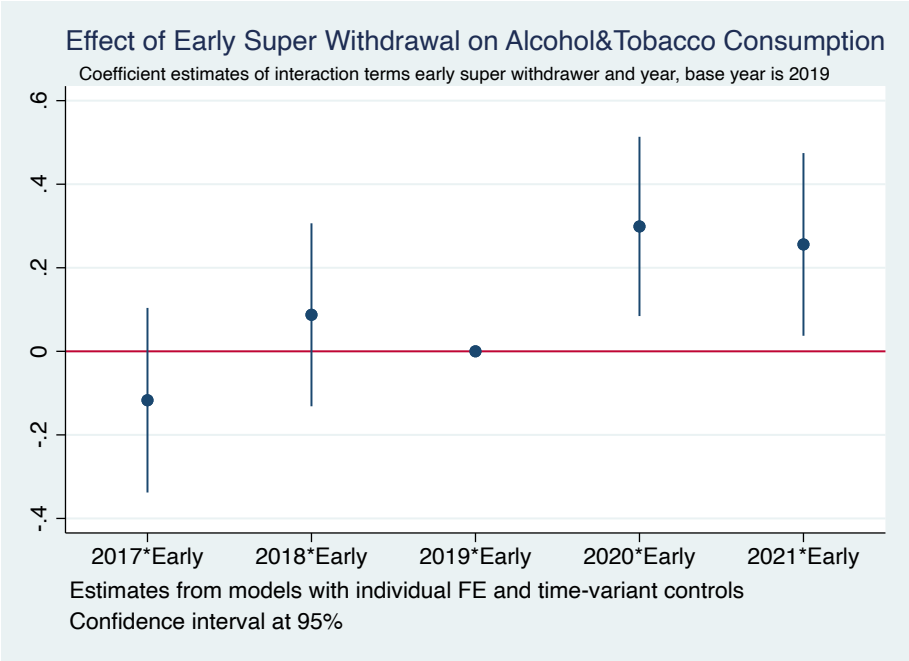
Graph 9 plots the coefficient estimates of interaction terms between the treatment group and year dummies, using 2019 as the base year. The estimates before the scheme was introduced in 2020 were a test for pre-parallel trends. All estimates before 2020 were insignificant, suggesting that the parallel trend assumptions were not violated. This gave me more confidence to give the results a casual interpretation. But passing the test of the prior trend did not mean that the parallel trend assumption is valid. The assumption would be more plausible if the treatment and control groups had similar characteristics, but people who had little to no super tended to earn less. Although my model included controls for observable characteristics such as income, some unobservable characteristics may be associated with consumption. This bias should be kept in mind when interpreting the results. The estimated

effect rests on the assumption that the observed trend in the eligible people with little super represented the counterfactual trend in the people who withdrew super.

The coefficient estimates of 2021*Early were highly significant. It is not surprising that the effect became obvious in 2021 even though the scheme was introduced in 2020 because of two reasons. Consumption opportunities were somewhat limited in 2020 due to lockdown restrictions. The restrictions were lifted in 2021, giving people more opportunities to spend. Some of the spending in 2021 could also reflect pent-up demand during the lockdown. In addition, people who applied in late 2020 might not receive the withdrawal until 2021 as it took some time for the applications to be assessed. My estimates suggest that withdrawing super early increased total spending by 6% and food spending by 8% in 2021. The effect on discretionary goods was particularly large. The estimates indicate that it increased spending on discretionary goods by around 11% in 2020 and 13% in 2021 for those who withdrew super early. The effect was smallest on essential goods spending, with the estimate weakly significant at around 3%.

I then explored the spending patterns on each category of non-durable goods spending and found that the effect was especially large on alcohol and tobacco consumption (Graph 10). The coefficient estimates of 2020*Early and 2021*Early were highly significant. The point estimates indicated that accessing superannuation early increased expenses on alcohol and cigarettes by around 30% in 2020 and 26% in 2021. Interestingly, the effect on alcohol and cigarettes was larger in 2020, whereas the effects on other types of spending tended to be larger in 2021. This could be because the lockdown restrictions in 2020 led people to drink more to cope with anxiety and boredom, and there was limited access to other types of entertainment activities.

Graph 10



The analysis could underestimate the true impact of the policies on consumption as it was limited to non-durables spending. Durables spending is more discretionary and could be more responsive to income shocks. Previous literature shows that the effect of wealth shock is larger for durables spending than non-durables spending (Price, Becker and La Cava 2019). As people were forced to spend more time at home due to the lockdown, household consumption of home entertainment units and appliances saw the largest growth during the pandemic (RBA, 2022 March bulletin). But the spending on these durable goods was not included due to data availability issues. Therefore, it is likely that the estimates represent the lower bounds of the impact of the policies on consumption.

Further, there is ample evidence that recessions typically have long-term scarring effects on employment and consumption. Consumption patterns might have been permanently shifted in the post-pandemic world. The level of risk aversion and incentives for precautionary

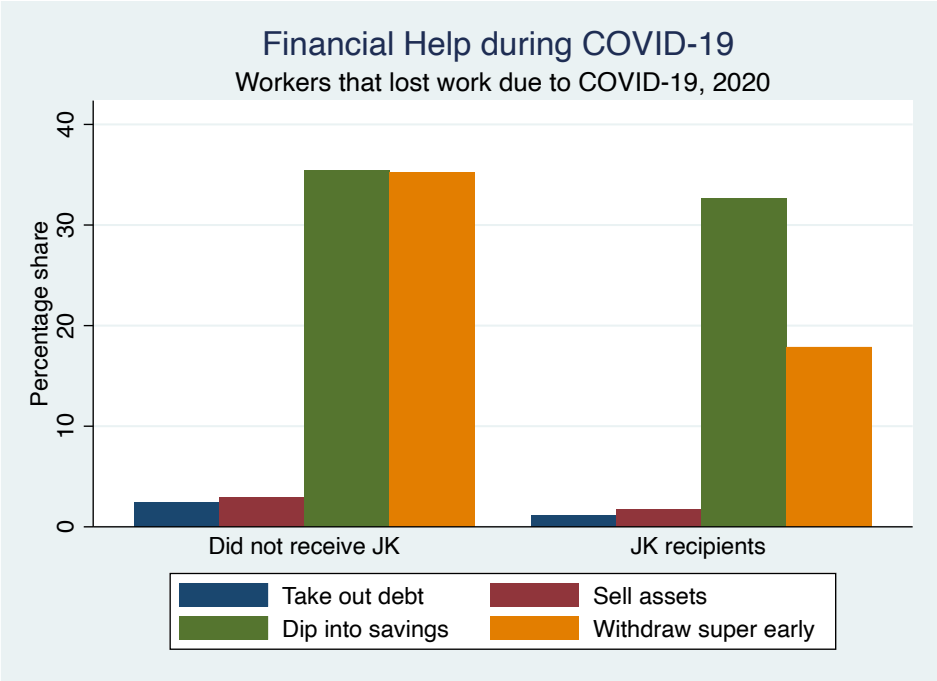
savings for different individuals might have changed. Some long-run impacts might have not yet flown through to the 2021 data. Lastly, it is not possible to completely disentangle the effect of the stimulus payments from other simultaneous shocks in the economy.

Impact of the JK Payments on Financial Stress Experience

Descriptive evidence

Some workers who lost their jobs as a result of the pandemic were not eligible to receive the JK subsidy. They might have to seek financial help in other ways. The special COVID-19 theme data in 2020 show that for people who lost work due to COVID-19, the JK recipients were less likely to seek other types of financial assistance than others (Graph 11). Most noticeably, the non-JK recipients were twice as likely to withdraw super early as the JK recipients. The non-JK recipient's group mainly consisted of casual employees who failed the 12-month job tenure requirement. This adds weight to a common criticism of the JK program which argues that short-term casual employees should have been included in the JK program. Withdrawing superannuation early meant that these casual workers would lose out on the benefits of compound interest, further reinforcing inequality.

Graph 11

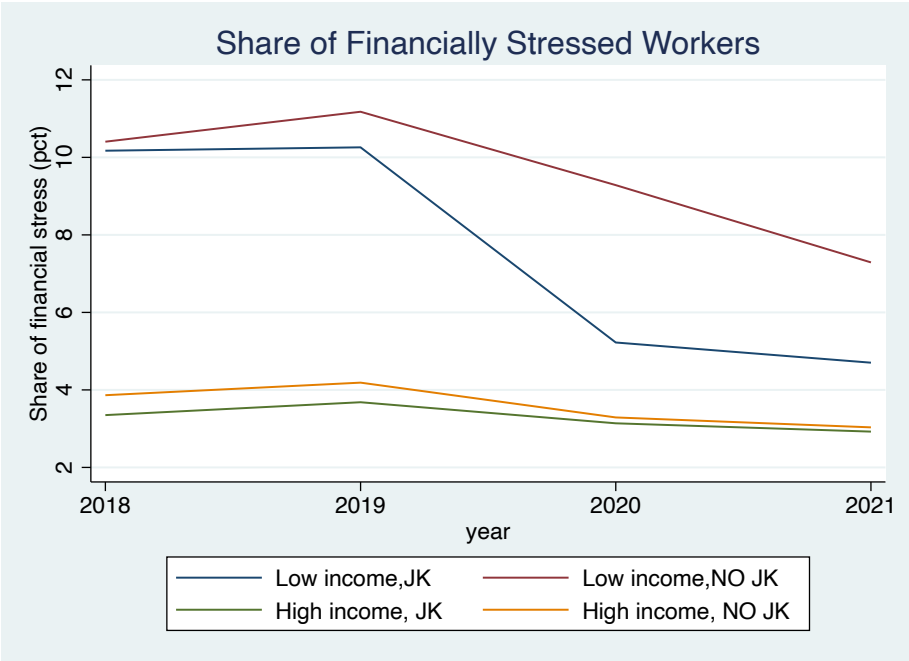


Next, I explored the panel data on financial stress events from the regular questions that were asked every year. Following the convention of HILDA data research, I defined financially stressed people as those who experienced 3 or more financial difficulties in the survey year. This does not necessarily capture households in deep financial trouble, it could reflect a short-term liquidity problem.

Graph 12 plots the average shares of workers experiencing financial stress by income level and whether they received the JK subsidy. Only people who were employed before the pandemic were included. The data show that the share of workers experiencing financial stress has dropped significantly in 2020 for low-income workers, especially for the JK receivers. This provides descriptive evidence that the JK payment is correlated with less financial stress experiences for low-income workers. High-income workers were a lot less likely to experience

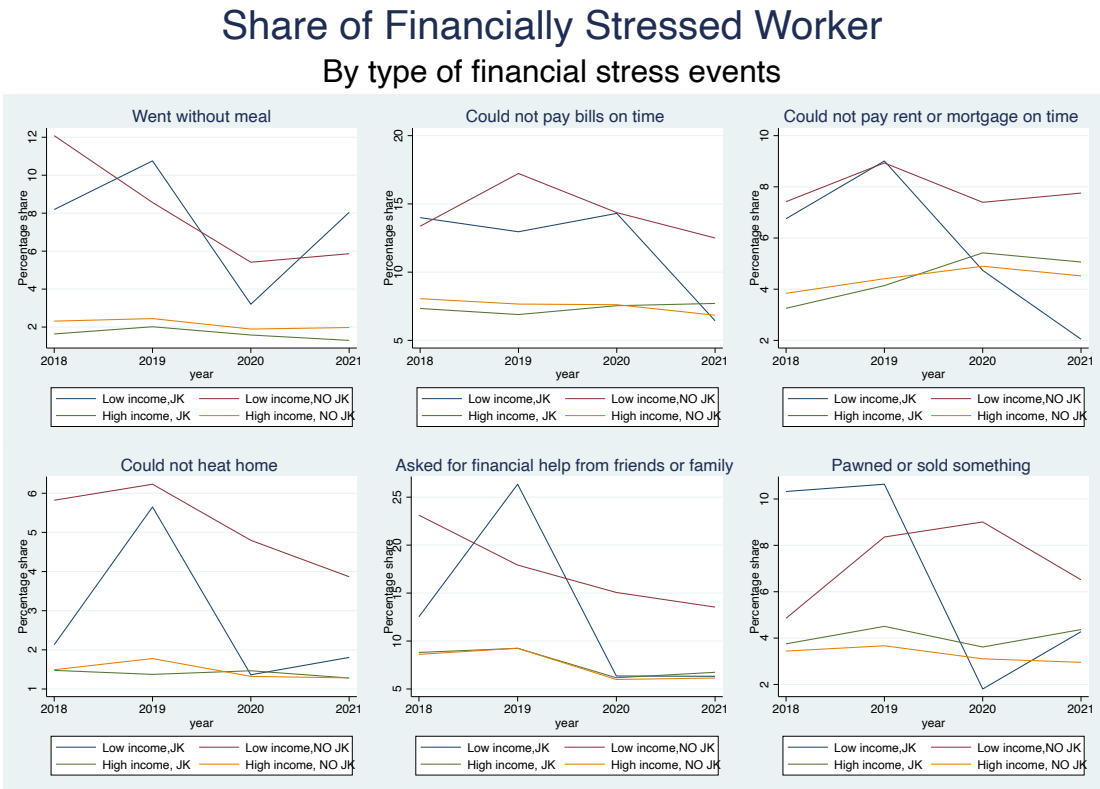
financial stress than low-income workers, and the trends were similar for those who received or did not receive the JK payments.

Graph 12



Breaking down the financial stress events by type, low-income workers who received the JK payments became less likely to encounter almost all types of financial difficulties after the JK program was introduced in 2020 (Graph 13). In particular, the declines in shares of people who could not pay rent or mortgage on time or had to sell something due to financial difficulties were most obvious for the low-income JK receivers.

Graph 13



Regression evidence

To formally examine the causal effect of the JK payments on financial stress for low-income employees. I used a difference-in-difference approach. The baseline model is specified as:

$$Financial\ stress_{it} = \alpha_i + \sum_t \beta_t (Treatment_i \times Year_t) + \sum_i \delta_i\, Treatment_i + \sum_t \phi_t Year_t + \gamma Controls_{it} + \varepsilon_{it}$$

The dependent variable $Financial\ Stress_{it}$ is a dummy variable that represents an individual who experienced 3 or more financial stress events in the past year. As a robustness check, I experimented with defining financial stress as experiencing 2 or 4 financial stress events. The results were broadly consistent with the baseline definition.

The coefficient of the interaction term between time and treatment dummies β captured the causal effect of the JK payments on financial stress experience. The β estimates were negative and insignificant. This suggests that the effect of the JK program on financial stress after controlling for other variables was negligible.

Overall, the data show that receiving the JK payments was associated with experiencing less financial stress for low-income households, but it does not appear to be a driver of reducing financial difficulties after controlling for other factors. People who lost jobs and could not receive the JK payments were much more likely to withdraw superannuation early and accessing the ERS was associated with less financial stress experiences. Receiving a large lump sum of money could alleviate short-term liquidity problems.

Conclusion and Discussion

The essay uncovers the heterogeneity in outcomes hidden in aggregate data and sheds light on the distributional effects of the JK payments on financial stress and consumption. It provides some evidence that the JK payments boosted the consumption levels of low-income workers. Receiving the JK payments was associated with less financial stress experiences for low-income workers. But no regression evidence supports the idea that it prevented households from experiencing financial difficulties after controlling for other factors. The findings contribute to several aspects of the discussion on the design of government policies.

The flat payment feature of the JK program raised a heated discussion on equality vs. equity. Job retention programs that offered indexing payments such as the UK's Furlough scheme (which can replace at most 80% of workers' regular income) might have increased inequality. Gortz, McGowan, & Yeromonahos (2021) found that furloughed employees

reduced their consumption and had a higher chance of experiencing housing stress among renters as a result of the reduced income. In contrast, the JK program was shown to close the inequality and poverty gaps (Borland & Hunt, 2021 Phillips, Gray, & Biddle, 2020). My findings that the JK program benefited workers from low-income households more provide further evidence that the flat payment structure has helped to reduce inequality.

The heterogeneous effect of the JK program implies that short-term casual workers who were excluded from the program would have benefited the most from the program as they typically have low incomes. This supports the argument that short-term casual workers should have been included to ensure the program is more targeted to people who needed the most payment. Around 40% of casual workers did not meet the requirement of being employed with the same worker for at least 12 months. These people were disproportionately more likely to work in industries that were forced to shut down during the pandemic. It was unfair to exclude them from the JK program. Although these workers were eligible to receive other government benefits, other payments were less generous than the JK. Furthermore, the heterogeneity in people who benefited from the program might imply that the payments to high-income workers were excessive as they did not need the cash transfer as much as others. This all boils down to a discussion of equality vs. equity. It is important to note that the primary goal of the JK program was to maintain employer-employee relationships, not to stimulate consumption or reduce financial hardships directly. Also, it is not possible to evaluate the counterfactuals. We would not know what the economy would look like without the generous JK support payments.

Further, I show that the ERS scheme boosted consumption substantially. But the results should be interpreted with caution as they are subject to the validity of the parallel trend assumption. The ERS could be a useful mechanism to provide support for temporary income

loss. It is a much more cost-effective policy response than a wage subsidy program as the superannuation was financed from individuals' superannuation accounts. But it is important to bear in mind that the reductions in retirement savings as a result of withdrawing super early could have long-run implications beyond this essay's scope.

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Appendix

Table A1: Effect of Job Keeper Payment on Consumption
Low income

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Dependent variable	Non-Durable	Non-Durable	Non-Durable	Non-Durable	Food	Food	Food	Food	Essential	Essential	Essential	Essential	Discre-tionary	Discret-ionary	Discre-tionary	Discret-ionary
Panel A: Estimates from basic DID model																
JK*after2020	0.110**	0.0942*	0.101**	0.0886*	0.0932	0.0517	0.0911	0.0491	0.122**	0.0916	0.114**	0.0878	0.0792	0.147*	0.0789	0.156**
	(0.0521)	(0.0539)	(0.0509)	(0.0527)	(0.0574)	(0.0541)	(0.0553)	(0.0524)	(0.0567)	(0.0577)	(0.0542)	(0.0557)	(0.0811)	(0.0778)	(0.0809)	(0.0774)
Panel B: Estimates from dynamic DID model																
2017#JK	-0.0205	-0.0217	0.00508	-0.00545	0.0393	0.0434	0.0643	0.0539	-0.0392	-0.0497	-0.00769	-0.0385	-0.00385	0.0527	0.0361	0.0761
	(0.0803)	(0.0813)	(0.0755)	(0.0753)	(0.0774)	(0.0762)	(0.0743)	(0.0713)	(0.0831)	(0.0819)	(0.0766)	(0.0752)	(0.110)	(0.108)	(0.109)	(0.104)
2018#JK	-0.00485	-0.0107	0.00111	-0.00806	0.0662	0.0596	0.0613	0.0531	-0.0272	-0.0364	-0.0192	-0.0359	-0.00653	-0.00130	-0.00555	-0.0117
	(0.0662)	(0.0659)	(0.0638)	(0.0633)	(0.0713)	(0.0670)	(0.0678)	(0.0640)	(0.0738)	(0.0714)	(0.0706)	(0.0686)	(0.0981)	(0.0987)	(0.0984)	(0.0983)
2020#JK	0.0892	0.0807	0.0862	0.0711	0.0963	0.0703	0.0885	0.0541	0.0812	0.0685	0.0802	0.0517	0.0699	0.157*	0.0679	0.162*
	(0.0617)	(0.0600)	(0.0652)	(0.0607)	(0.0654)	(0.0587)	(0.0660)	(0.0594)	(0.0659)	(0.0645)	(0.0656)	(0.0634)	(0.101)	(0.0918)	(0.105)	(0.0928)
2021#JK	0.131**	0.0910	0.120*	0.0808	0.161**	0.123*	0.158**	0.114*	0.140**	0.0891	0.129*	0.0798	0.0840	0.103	0.0894	0.102
	(0.0654)	(0.0675)	(0.0672)	(0.0673)	(0.0721)	(0.0679)	(0.0698)	(0.0635)	(0.0680)	(0.0677)	(0.0706)	(0.0683)	(0.107)	(0.104)	(0.107)	(0.103)
Control variables	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y
Individual FE	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y

Robust standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1; Control variables include age, having young children and major life events in the past year